

Contact

Bangalore
+919743467537 (Mobile)
sudhir.13singh@gmail.com

www.linkedin.com/in/sudhir13s
(LinkedIn)

Top Skills

Jupyter
Agentic AI Development
AI Agents

Languages

Hindi (Native or Bilingual)
English (Full Professional)

Certifications

Spring Boot 2.0 Essential Training
Machine Learning Foundations:
Linear Algebra

Honors-Awards

PD RockStar
Pixel Award
Seek It Award
Team up to Excel
Philips HSDP hackathon

Sudhir Kumar Singh

Relocated back to India | Transitioning to AI/ML | Generalist Software Engineer

Bengaluru, Karnataka, India

Summary

I'm an enthusiastic learner and a software generalist (trying to be one actually) who is fascinated by learning and solving problems. I am a passionate and versatile developer who enjoys solving complex problems and designing innovative solutions. My interest ranges from security to networks to machine learning and robotics. I believe AI, robotics, and the cloud will transform into a new era and solve many of the existing problems we have today.

Currently, I am working with Walmart Global Tech in their Eligibility and Restriction team in a Senior Engineer capacity, building microservices to derive the eligibility and restrictions of an offer, item, or product at the offer or store level. Prior to this, I was a Software Engineer III at Strateos Inc., a startup that provides a cloud-based platform for automating scientific labs and significantly accelerating drug discovery. There, I developed and integrated various services and components using Java, Spring Boot, Scala, Akka, and Python, leveraging microservices and event-driven architecture. I also led multiple product feature epics from inception to production, collaborating with cross-functional teams and stakeholders.

Experience

Walmart Global Tech

Senior Software Engineer

November 2023 - Present (2 years 6 months)

Sunnyvale, California, United States

- Building microservices responsible for deriving eligibilities and restrictions of an offer, item, or product at the offer, store, and fulfillment center level for all markets (Java17, WebFlux, Kafka, Cassandra, k8s)

- Collaborated with the data science team to integrate predictive model results to optimize delivery time to improve On-Time Delivery for 3rd party sellers' products and deployed scalable service to production.

- Implemented eligibility feature to enable cross-border shopping for customers on walmart.com for Walmart or 3rd party sellers items.
- Improved internal automation framework tool (based on Intuit Karate) to run regression test suites (UI/Microservices/Kafka/Database) in parallel and reduced execution time by ~60%. (Java8, Kotlin, JS)

Strateos

Software Engineer III

July 2021 - June 2023 (2 years)

Menlo Park, CA

- Developed LoadStar product by integrating with third-party automation solutions (Thermo Fisher Scientific Momentum, Hamilton Venus) to automate a scientific lab and significantly reduce drug discovery time (more than 6m-2y). (Python, Spring Boot)
- Designed and developed Asset-management service and UI solution (cloud lab) to orchestrate and manage lab, workcell, and devices in a laboratory. (Java, Spring Boot, React, ECR, EKS)
- Developed components using event-driven architecture to compile a scientific intent into device or robot commands; schedule, run, and report the execution of commands. (Scala, Akka, C#, RabbitMQ)
- Led many product feature epics from their inception to production deployment and everything in between. (Colony picking, device migration to cloud lab)
- Worked with many device drivers to automate command execution (liquid handle, incubation, seal) and supported 30+ vendor hardware and robots (Tecan Omni, PreciseFlex 400). (Scala, Akka, RabbitMQ)
- Worked with complex ansible scripts to provision a NUC and deploy services including installation and configuration of complex software such as virtual box and installation of Windows & dependant software/libraries in it).

Center On Access Technology, RIT/NTID

ML Software Engineer

January 2021 - May 2021 (5 months)

Rochester, NY

- Building language and error models on data received from Automatic Speech Recognition systems to improve hybrid captioning in a hard-of-hearing classroom setting. (Python, Jupyter)
- Developing services and pipelines for deploying machine learning models on AWS. (Python, Flask, React, S3, DynamoDB, EB)

Advarra

Software Engineer Co-op

January 2020 - April 2020 (4 months)

Rochester, New York Metropolitan Area

- Improved page load in Clinical Conductor (Clinical Trial Management System) from ~4-5s to ~1-2s in ~10% of web pages by optimizing SQL queries (reduced joins and common table expressions) and simplifying core logic. (C#, ASP.NET MVC, MSSQL)
- Developed tools for internal stakeholders to manage clients' data in Pendo.io using their REST APIs. (C#)

Rochester Institute of Technology

Graduate Technical Assistant

January 2019 - December 2019 (1 year)

Rochester, New York Metropolitan Area

- Built and deployed a credential management portal to enable CS students to manage their credentials resulting in a reduced system admin workload by ~40%. (Python, Flask, jQuery, Apache2, LDAP, AD)
- Assisted students and faculty members of the CS department with their day-to-day activities and issues as a System Admin assistant.
- Installed, configured, and maintained CS department computer labs, servers (i.e. Mail server, Web server), research machines, and Raspberry Pis.
- Helped in the transition from LDAP to AD accounts, retrofitted internal tools, portals, and wiki.

Philips

Senior Software Engineer

February 2015 - July 2018 (3 years 6 months)

Bengaluru

- Developed reliable REST APIs for HSDP (a PaaS product) to enable internal and external clients to exchange patients' medical information using the FHIR and HL7 standards.
- Built a prototype and showcased HSDP capabilities as SMART on FHIR by integrating with HealthCloud (a Diabetes app), eCareCoordinator (a Salesforce monitoring app), EMR systems, and the internal data repository.
- Oversaw a team of 8 developers in the absence of a team architect and headed the implementation of the Oncology integration project for two releases.

Clinisys

3 years 9 months

Software Engineer

September 2012 - February 2015 (2 years 6 months)

Bengaluru

- Contributed to the implementation of Distributed Print Server; an application to allow printing of reports on registered network printers across different units of an organization. (C#.NET, MSSQL)
- Designed and developed key product features like Timeout - for application security and Auto-populate fields on barcode scan - for better accessibility for the pathology product. (Delphi, C#.NET, MSSQL)
- Supported product deployments and handled many highly escalated & business-critical client issues.
- Created many custom patient and financial report templates for clients using SAP Crystal Reports.

Associate Software Engineer

June 2011 - August 2012 (1 year 3 months)

Bengaluru

- Led a project with 4 developers and efficiently delivered & deployed 3 parallel product releases on time with zero clients reported defects along with product client support.
- Owned a product and was involved in end-to-end cycles from the requirement, design, development, deployments, and client interactions.

- Implemented many network interfaces (i.e. ADT, Orders, Results, and Scheduling) to exchange patient information using the HL7 standard over TCP/IP protocol.
- Designed and developed features like ICD10 and Meaningful Use 2 into the product to help clients to transition to the next healthcare leap mandated by the US government.

Education

Rochester Institute of Technology

Master of Science - MS, Computer Science · (2018 - 2021)

Rajiv Gandhi Prodyogiki Vishwavidyalaya

Bachelor of Engineering - BE, Information Technology · (2006 - 2010)